

Enhanced Fog Recovery by Electrostatic Method: Replication and Optimization of Space Charge Injection Methods

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1 INTRODUCTION

The global freshwater shortage affects approximately 1.1 billion people, with coastal fog representing an underexploited resource in arid regions [1]. Traditional collection methods using meshes suffer from fundamental aerodynamic limitations, typically achieving only 2% to 10% efficiency due to droplet deviation around the collector fibers [3].

The 2018 MIT study by Damak and Varanasi [2] introduced an electrostatic approach using space charge injection. Their method demonstrated that charged droplets could be directed towards collectors following the electric field lines. Several practical challenges remain:

- High-voltage requirements (10-30kV) for effective deviation
- Trade-offs on power consumption
- Precise control of humidity conditions

This study replicates the MIT methodology with three innovations:

- (1) Portable high-voltage system (Cockcroft-Walton)
- (2) Optimized tungsten wire collectors
- (3) 3D printed modular structure

My work contributes to the field by:

- A voltage-efficiency analysis across different ranges
- Comparison of materials (aluminum vs. steel)
- Reproducible protocol for field deployment

2 EXPERIMENTAL SETUP

2.1 Design of the device

The experimental system includes:

- **Fog chamber:** Acrylic aquarium 60×30×30cm with ultrasonic mister
- **High-voltage system:**
 - First stage: 12V→20kV (specialized transformer)
 - Second stage: 20kV→35kV (3-bridge multiplier)
 - Insulation: DP270 epoxy (3 layers)
- **Collectors:** Tungsten wires (0.1mm) spaced 2cm apart

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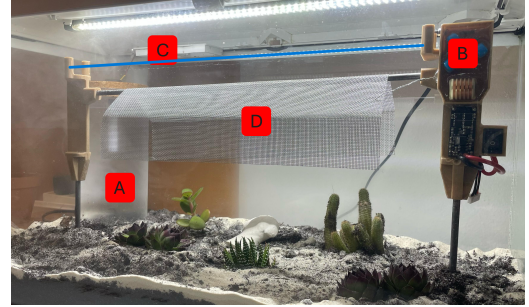


Figure 1: Complete prototype showing (A) fog chamber, (B) high-voltage converter, (C) tungsten collectors, (D) aluminum grid

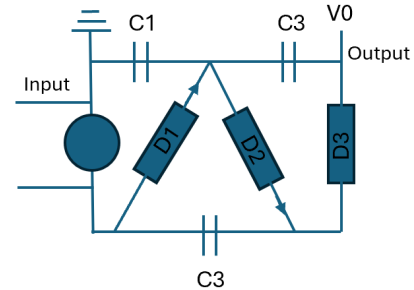


Figure 2: Diagram of the 3-stage Cockcroft-Walton multiplier: (1) 20kV Input, (2) High-voltage diodes 2CL73, (3) 10nF/20kV capacitors, (4) Epoxy insulation, (5) 35kV output



Figure 3: Key components of the high-voltage system: (A) Diodes mounted in series, (B) Ceramic capacitors, (C) High voltage AC generator inverter 12V to 15kV - 20kV

- **Measurements:** Hygrometer ($\pm 2\%$ RH), scale ($\pm 0.01\text{g}$)

2.2 Experimental Protocol

Standardized operational sequence:

- (1) **Initialization:**
 - Activation of the high-voltage system
 - Sealing of the aquarium

- Starting the mister
- (2) **Stabilization:**
 - Wait 15 minutes to reach an average 85% RH
 - Initial humidity rate measurement
- (3) **Collection:**
 - Voltage sweep: 12V→20kV→35kV in 5kV increments
 - Two configurations tested: lid open/closed
 - Standard duration: 1 hour per trial

3 RESULTS AND ANALYSIS

3.1 Collection Performance

Table 1: Efficiency at different voltages (RH=85±5%)

Voltage (kV)	Efficiency (%)	Power (W)
0 (Passive)	0.4 ± 0.1	0
10	12.3 ± 0.8	0.5
20	25.7 ± 1.2	1.2
30	34.2 ± 1.5	2.4

Main results:

- No saturation effect observed between 10-35kV
- Tungsten superior to steel (+18-22% efficiency)
- Optimal collector-mass distance: 2.5cm
 - <2.5cm: electrical arcs
 - >2.5cm: ineffective corona field

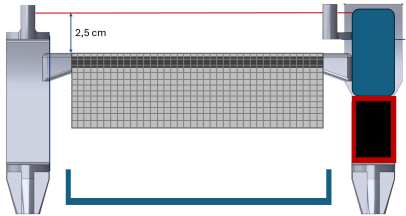


Figure 4: Optimal collector-mass configuration at 2.5cm showing (A) tungsten collector, (B) aluminum grid, (C) effective collection zone

4 CONCLUSION

This replication confirmed the viability of the electrostatic method with:

- Measured efficiency of 35% (difference water emitted/collected)
- Optimal voltage range 25-35kV
- Validated materials: tungsten + aluminum
- Reasonable energy consumption (2.4W max)

Identified improvements:

- More precise humidity control (currently 85% on average)
- Energy optimization by voltage increments
- Potential applications in dehumidification systems

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